Bubble story numbers

Soap Bubbles

Name _____ Date _____

Use these story numbers about soap bubbles. The picture is not a ruler.

1. Little Wonderer blows 28 bubbles, then 17 more float up. How many bubbles is that altogether?

28 + 17 = _____

Show your thinking.

2. Big Wonderer counts 64 wand dips in the jar, then uses 29 of them. How many dips remain?

64 − 29 = _____

Show your thinking.

3. The little one sketches 36 color stripes, then adds 48 more stripes after the film thins. How many stripe marks are in the notebook?

36 + 48 = _____

Show your thinking.

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4. A tray holds 81 floating bubbles. Then 37 pop in the breeze. How many bubbles are still floating?

81 − 37 = _____

Show your thinking.

5. They time 19 short-lived bubbles and 56 longer ones. How many bubbles did they time in all?

19 + 56 = _____

Show your thinking.

6. A bottle starts with enough mix for 90 bubbles. After play, 38 bubble turns are used. How many turns of mix are left?

90 − 38 = _____

Show your thinking.

A first bubble

One soft blow lifts a round soap bubble into the light. Its thin wall already hides a quiet science story.

Thin soap film

Water mixed with soap makes a thin shiny film. Air rests safely inside that delicate watery wall.

Colors that swirl

Light bounces inside the thinning soap film. Soft color stripes slide as the wall slowly changes.

Stripes to notice

Sketch the moving color stripes in a notebook. They hint that the soap film is getting thinner.

Why so round?

Surface tension pulls the soap film evenly from every side. A free bubble settles into a round sphere.

A careful pop

A hole in the thinning film ends the bubble at last. Tiny droplets scatter softly through the light.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

Soap Bubbles

Give the morning your words.

Soap Bubbles · Curiosity day

Name _____ Date _____

Choose a bubble scene. Look, say an idea, then draw or write.



A first bubble



Colors that swirl



A careful pop

Draw a bubble wall. Label the film and the air inside. Mark where colors swirl.

soap film · interference · surface tension · evaporate · sphere · because

At first, … what lifts into the air?

Then the colors, … what makes them shift?

When it pops, … what happens to the film?

Read your words aloud. Did you connect a clue (color, shape, or pop) to an explanation?



Please Don't Poke the Shine

Little Wonderer blew a soft silver ring,
and a round bit of daylight rose up on a string
of invisible air. "It's a jewel!" said the small one.
Big Wonderer smiled. "It's a film — watch it run."

Colors slid over it, quiet as oil on a spoon,
greens into pinks in the thin afternoon.
"Is it painted?" the little one asked with a grin.
"No paint," said the big one. "Light bouncing within."

They sketched every stripe as the shimmer grew wide,
then asked why a free bubble wears a round side.
"The film pulls in evenly, soft as a sigh —
a sphere is the thrifty shape under the sky."

A hush, then a glitter of droplets like rain.
The shine had let go. Only air would remain.
So if you find rainbows that float for a while,
look kindly, keep poking for never — just smile.

Soap Bubbles

01 What is a soap bubble made of?



A watery soap film wraps a pocket of air.

Look closer. Find the thin shiny wall of the bubble, and the air space inside.

A little explanation. A soap bubble is a very thin film of water mixed with soap, wrapped around a pocket of air. Soap molecules help the watery film hold together a little longer than plain water would. The wall is so thin you can see through it, yet it still encloses the air.

Try this. Dip a wand and hold up a flat soap film (not a bubble yet). Look at the film edge-on if you can, then blow gently to trap air inside.

02 Why do bubble colors swirl and shift?



Light and a thinning film make colors shift.

Look closer. Find places on the bubble where stripes of color move and change.

A little explanation. Light reflects from the front and back of the thin soap film. Those reflections can add or cancel, so some colors look bright while others fade. As water drains or evaporates, the film gets thinner, and the color pattern shifts. Scientists call this thin-film interference.

Try this. Watch one bubble without poking it. Name two colors you see, then notice whether they move toward the top or bottom as time passes.

03 Why are free-floating bubbles round?



Surface tension pulls a free bubble into a sphere.

Look closer. Compare a free bubble in the air with any stretched film still on a wand.

A little explanation. Surface tension pulls on the soap film from every direction. For a given amount of air inside, a sphere uses the least film area, so a free bubble settles into a round shape. On a wand, the frame can stretch the film into other shapes until the air is trapped.

Try this. Blow one bubble and watch it from the side. Sketch the round outline, then write one sentence about the pull of the film.

04 Why does a bubble pop?



When the film breaks, the bubble is gone.

Look closer. Find a dark or dull patch on a bubble just before it vanishes, if you can.

A little explanation. A bubble pops when its film breaks. Touching it, dusty air, or a dry spot can open a hole. Even without a poke, water leaves the film as it drains and evaporates, so a place can get too thin to hold. Some recipes last longer, but every ordinary soap bubble still has a short life.

Try this. Time how long two bubbles last in still air. Change only one thing (shade versus sun, or a gentle cover) and compare. Do not aim bubbles at faces or eyes.